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Review

Effect of GVHD on the gut and intestinal microflora

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ABSTRACT

Graft-versus-host disease (GVHD) is one of the most important cause of death in patients undergoing allogeneic hematopoietic stem cell transplantation (allo-HSCT). The gastrointestinal tract is one of the most common sites affected by GVHD. However, there is no gold standard clinical practice for diagnosing gastrointestinal GVHD (GI-GVHD), and it is mainly diagnosed by the patient's clinical symptoms and related histological changes. Additionally, GI-GVHD causes intestinal immune system disorders, damages intestinal epithelial tissue such as intestinal epithelial cells((IEC), goblet, Paneth, and intestinal stem cells, and disrupts the intestinal epithelium's physical and chemical mucosal barriers. The use of antibiotics and diet alterations significantly reduces intestinal microbial diversity, further reducing bacterial metabolites such as short-chain fatty acids and indole, aggravating infection, and GI-GVHD. gut microbe diversity can be restored by fecal microbiota transplantation (FMT) to treat refractory GI-GVHD. This review article focuses on the clinical diagnosis of GI-GVHD and the effect of GVHD on intestinal flora and its metabolites.

1. Introduction

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) can kill residual malignant cells by donor T cells to treat hematological malignancies [1]. However, life-threatening complications, including graft-versus-host disease (GVHD), infections, and disease recurrence, may occur after the transplantation. GVHD is characterized by host tissue and organ damage mediated by donor immune cells; it is highly morbid and increases mortality [2]. The skin, liver, and gastrointestinal tract (GIT) are the most common tissues affected by GVHD. Gastrointestinal GVHD (GI-GVHD) is one of the most difficult to treat and the main cause of death after allogeneic stem cell transplantation. GI-GVHD causes intestinal epithelial tissue damage by activating the immune system of allogeneic grafts, especially intestinal stem cells (ISC), Paneth cells, and goblet cells [3], producing multiple clinical symptoms and digestive endoscopic and histological changes. The homeostasis of intestinal microbial diversity and related metabolites are markedly linked with the outcome of allo-HSCT and the development of GVHD [4]. GVHD incidence and antibiotics use causes the loss of intestinal bacterial diversity and the changes in the metabolites produced by bacteria [5]. In addition, it has been shown that restoring gut microbiota diversity through fecal microbiome transplantation (FMT) facilitates the treatment of refractory GI-GVHD.This review focuses on the manifestations of GI-GVHD and the changes in the gut microbiota to provide in-depth knowledge for future clinical research and application.

2. Effect of GVHD on the digestive tract after hematopoietic stem cell transplantation

GVHD is a major cause of morbidity and non-recurrence mortality

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(NRM) after allo-HSCT, and gastrointestinal (GI) complications are one of the most frequent manifestations of GVHD. GVHD can manifest in any part of the GIT; it is most common in the cecum, ileum, and ascending colon and may also involve the stomach, duodenum, and rectum [6]. Clinically, it is most commonly characterized by nausea, vomiting and anorexia, abdominal pain, and weight loss. Involvement of the lower digestive tract indicates severity, as manifested by diarrhea with or without hematochezia and abdominal colic. In severe cases, diarrhea is watery and even bloody [7]. GI endoscopy and histological examination are crucial for diagnosing GI-GVHD when suspected based on the patient's symptoms. Endoscopic appearance changes such as a mucosal crypt, gland expansion, and crypt abscess or occlusion have reference significance for diagnosing GI-GVHD [6]. The typical pathological features of epithelial cell apoptosis, recess / gland apoptosis, shedding, and ulceration can further clarify the diagnosis [8].Furthermore, the diagnosis accuracy can be improved by ultrasound [9],Computed Tomography [10,11], magnetic resonance enterography [12],and Positron emission tomography–computed tomography [13]. However, currently, there is no gold standard procedure for diagnosing GI-GVHD [6].

2.1. Effect of GVHD on the upper gastrointestinal tract

In upper GI-GVHD, The clinical manifestations of esophageal GVHD are usually dysphagia or chest pain [14]. Endoscopic esophageal GVHD is characterized by mucosal inflammation, epithelial cell atrophy, erythema, and erosion [15]. apoptosis and increased intraepithelial lymphocytes were diagnostic features [16]. The initial clinical manifestations of gastroduodenal GVHD may be occult'. Endoscopic findings range from mild mucosal erythema and edema to obvious erosion, ulceration, atrophy, and mucosal exfoliation, which may be an effective diagnostic indicator for upper GI-GVHD [17]. Although gastric mucosal exfoliation is uncommon under endoscopy, its specificity and positive predictive value (PPV) are 100% [18]. Gastric biopsy revealed that the most common gastric GVDH manifestations were apoptosis, gland destruction, sparse inflammatory infiltration, and granular eosinophil debris [8]. There is a close correlation between histological and clinical grades of GVHD in the upper GIT, where a clinical grade of GVHD has higher NRM predictive accuracy [19].

2.2. Effect of GVHD on the lower digestive tract

Intestinal GVHD progresses to life-threatening symptoms such as protein loss enteropathy, hypoalbuminemia, bloody diarrhea, and intestinal obstruction [20]. Endoscopic manifestations such as villous atrophy, redness, edema, erosion and shallow and wide ulcerative lesions in the intestine have diagnostic significance for GVHD in the lower gastrointestinal tract,Although mucosal exfoliation is uncommon, it significantly predicts severe GVHD [21]. The specificity of biopsy was higher relative to endoscopy and better associated with histological grade or clinical severity [6,22]. The histological examination can diagnose GVHD lesions in mucosal areas that appear to be normal under endoscopy [6]. Apoptosis in the crypt with a reduction in the number of lamina propria lymphocytes is the most well-characterized histological manifestation [23]. Furthermore, mucosal ulcers and atrophy, glandular dilatation, focal fibrosis, and crypt abscess can occur, too [24]. Histopathological examination plays a key role in diagnosing and grading GVHD in various intestine regions.

3. Protective effect of intestinal epithelial cells against GI-GVHD

Intestinal epithelial tissue mainly includes different functional IECs, ISCs, goblet cells, and Paneth cells. IECs are primarily responsible for nutrient absorption and provide a physical barrier through tight junctions to separate the intestinal wall from the lumen. Secretory cells such as Paneth and goblet cells maintain intestinal microbial ecology, homeostasis between intestinal flora and immune response, and protect the host from pathogens [25,26]. ISC located at the base of the crypt can produce all cell types of intestinal epithelium.

The goblet cells protect IEC by forming an intestinal physical barrier through tight junctions and secreting mucus [25,27]. The main component of the intestinal mucus is the glycoprotein MUC2, a dense mucosal barrier that is difficult for most intestinal bacteria to penetrate [28]. However, it is consumed by intestinal bacteria containing MUC2 glycan metabolic enzymes. Therefore, alterations in the intestinal bacterial composition affect the integrity of the mucus layer [29]. The injury of goblet cells and intestinal flora alterations in GVHD patients disrupt the intestinal mucosal barrier, allowing the pathogens to translocate and transmit, causing a vicious cycle of GVHD and infection.

Paneth cells located at the crypt base of the small intestine construct a chemical/physical barrier in the intestinal mucosa to prevent microorganisms from invading the crypt environment [30]. Various antimicrobial peptides (AMPs), including intestinal α-defensins, lysozyme and secretory phospholipase A2 (PLA2), and regenerative islet-derived protein 3α (REG3α), are secreted in the intestinal lumen upon sensing bacteria and bacterial metabolites to maintain intestinal microbial diversity and homeostasis [31]. It also acts against pathogenic bacteria, fungi, and some viruses [31]; however, its activity against symbiotic bacteria is weak. Therefore, it can protect the host from intestinal pathogens by regulating the composition of intestinal microbiota [32,33]. Paneth cells also maintain ISC activity by expressing EGF, TGFα, Wnt3, and Notch ligands [34], making them a 'crypt guardian' [35](Fig. 1). In intestinal GVHD, the degree of Paneth cell destruction has been observed to be associated with the severity of GVHD and non-recurrent death [26]. Paneth cell destruction causes a reduction in AMP secretion, thereby promoting intestinal flora imbalance and pathogenic intestinal bacteria transformation [36]. It also promotes microorganisms to destroy the mucosal epithelial barrier and activate antigen-presenting cells, further leading to crypt damage and aggravation of GVHD [37].

Interspersed between Paneth cells and located at the base of the crypt is ISC, capable of producing all cell types of the intestinal epithelium tissue, essential for intestinal homeostasis and regeneration after injury. However, ISC can be used as the main cellular target of intestinal GVHD [38]. Their injury may be linked with the pathophysiology of intestinal GVHD, and strategies to protect them may be beneficial.

The damage of Paneth and goblet cells promotes changes in the intestinal microbiome and aggravate GVHD, causing pathogen-associated molecular patterns (PAMPs) to cross the intestinal barrier and activate the immune system. Monocytes can recruit macrophages and neutrophils to inflammatory sites by promoting Th17 cell differentiation [39]. Neutrophils cause damage to the intestinal barrier and tissues by generating reactive oxygen species and then migrate to mesenteric lymph nodes to participate in allogeneic antigen presentation [40,41]. Activated allogeneic donor T cells promote GVHD development (Fig. 2). Congenital lymphocytes (ILC) near the crypt play a role in an innate manner [42]; they lack antigen-specific receptors or antigen-specific responses, where ILC2 and ILC3 are located in the lamina propria of the small intestine and colon [43]. Intestinal ILC3 expresses major histocompatibility complex (MHC) class II, promotes tolerance to symbiotic bacteria, prevents excessive T cells activation during intestinal infection, and limits intestinal immunity and inflammation by presenting symbiotic antigens to T cells without providing co-stimulation [44–46]. In addition, ILC3 is a tissue-resident cell that produces interleukin-22 (IL-22) secreting Th17 family cytokines that limit the intestinal GVHD development, stimulate Paneth cells to secrete REG3α and promote ISC-mediated intestinal epithelial regeneration [47,48]. ILC3 can also maintain ISC function after tissue injury [49]. Therefore, ILC3 essentially regulates inflammation during GVHD. ILC2 improves allo-HSCT recipients' survival rate [50,51]. However, when GVHD occurs in the intestine, ILC2 decreases in the intestinal lamina propria. Infusion of exogenous ILC2 can reduce the activation of donor pro-inflammatory Th1 and Th17 cells. In addition, ILC2 can also promote the increase of donor myeloid-derived suppressor cells (MDSC) in the intestine by

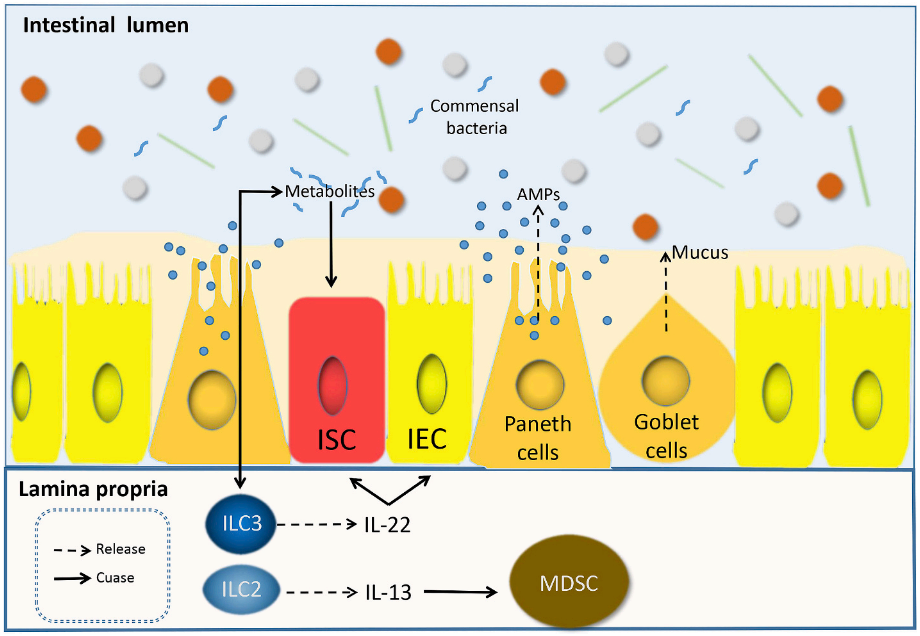

Fig. 1. Under physiological conditions, the host immune system maintains the diversity of the gut microbiota and prevents pathogens' growth. The surface of intestinal epithelial cells (IEC) maintains a complete barrier to prevent bacterial translocation in the host tissues. Paneth cells secrete antimicrobial peptides (AMPs) into the intestinal lumen to prevent IECs from contacting intestinal microbiome bacteria and potential pathogens. AMPs also maintain intestinal microbial homeostasis and provide nutritional signals for intestinal stem cells (ISC). Goblet cells consolidate the barrier of separating microbiota and host tissue by secreting mucin, thereby further protecting the intestinal epithelium from lumen bacteria. In addition, symbiotic bacteria stimulate innate lymphocytes (ILC) 3 in the lamina propria to produce IL-22, which stimulates Paneth cells to secrete AMP and maintain ISC. Metabolites of symbiotic bacteria can also reduce IEC injury and GI-GVHD by regulating intestinal inflammation and immunity. ILC2 promotes the increase of donor myeloid-derived suppressor cells (MDSC) by producing IL-13.

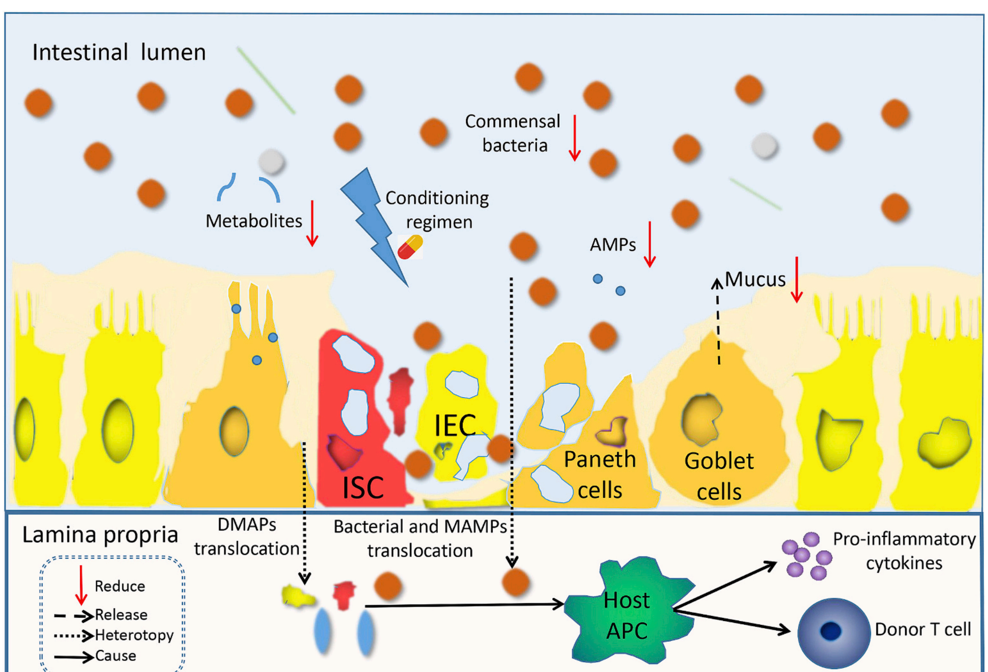

Fig. 2. Cytotoxic preconditioning regimen and GI-GVHD lead to intestinal cell damage. IEC damage causes intestinal barrier disruption. Goblet cell injury causes the decreased function of the intestinal mucosal barrier. Paneth cell injury leads to decreased AMPs secretion and loss of ISC niche function, and intestinal ecological imbalance. Antibiotic treatment and limited nutrients in the intestine also promote intestinal ecological disorders and further aggravate intestinal damage. Allowing the translocation of DAMP, bacteria, and MAMP to the lamina propria and activates host APC through TLR and NLRP3 inflammasome, causing pro-inflammatory cytokines release, donor T cells activation, and GVHD aggravation. Donor T cells and inflammatory cytokines (such as TNF, IL-6, and IL-1) destroy the ISC niche system.

producing IL-13. Therefore, ILC can protect the intestine from pathogens, promote intestinal lymphoid tissue development and anti-inflammatory and immune response, and help repair damage, thereby supporting intestinal homeostasis and reducing GVHD damage to the GIT [52,53].

4. Changes of intestinal flora after hematopoietic stem cell transplantation

The microbiota balance essentially regulates the intestinal immune system in vivo. Balanced gut microbiota can also protect the host from viruses, bacteria, and fungi by preventing GIT colonization of pathogens [54,55]. The literature suggests that removing gut microbiota reduces the risk of GVHD [56]. however, more in-depth research concluded with opposite results [57–59]. The imbalance of gut microbiota is closely related to the pathogenesis of GVHD during allo-HSCT [60]. The human gut microbiota comprises various bacterial phyla, of which Firmicutes Bacteroidetes, Proteobacteria, Verrucomicrobia, and Actinobacteria are the most common symbionts in the gut microbiota. The dominant obligate anaerobic bacteria are important in maintaining intestinal homeostasis [61]. The imbalance of gut microbiota in allo-HSCT patients is usually manifested by the loss of obligate anaerobic symbiotic bacteria, the pathogenic bacteria expansion, and decreased abundance and diversity of gut microbiota [62,63].

4.1. Changes of flora in Firmicutes

The amplification of *Enterococcaceae*, *Lactobacillaceae*, *Streptococcaceae* in *Lactobacillales*, and *Staphylococcaceae* in *Bacillales* is associated with the occurrence of GVHD [64,65]. Of these, *Enterococcus* and *Streptococcus* are the common dominant taxa [65]. The *Enterococcus* quantity significantly increased before or during GVHD after allo-HSCT [66]. Increased *Enterococcus* abundance (relative abundance ≥30%) was associated with increased GVHD risk and GVHD-related mortality in allo-HSCT recipients [67,68]. *Enterococcus* produces metalloproteinase, which aggravates intestinal inflammation by impairing epithelial barrier integrity and up-regulating the phosphotransferase system [69,70]. Furthermore, *Enterococcus* propagation requires lactose; therefore, a high lactose diet may cause higher GVHD morbidity and mortality [67]. These facts indicate the role of *Enterococcus* in GVHD pathogenesis. The decrease of intestinal *Lactobacillus* can aggravate intestinal GVHD. *Lactobacillus* can improve GVHD by regulating T helper cell 17 / T cell balance and limiting the expansion of *Enterococcus* [71,72]. However, when the abundance of *Lactobacillus* exceeds *Enterococcus*, it stimulates T cell reconstruction, including TH17, thereby promoting inflammation. Therefore, it is difficult to determine whether *Lactobacillus* expansion is completely beneficial [73].

Clostridia is the most abundant symbiotic bacteria in the intestine, and most of its members are Gram-positive anaerobic obligates, short-chain fatty acid producers [74]. *Lachnospiraceae* and *Ruminococcaceae* of *Clostridiales* can regulate acetylated H3 levels in CD4 + T cells and maintain the regulatory T cells (Treg) / Th17 ratio. Their metabolites (SCFs) can induce immune tolerance through Treg cells [75]. A higher abundance of *Lachnospiraceae* and *Ruminococcaceae* increases the number of natural killer (NK) and B cells, alleviates GVHD symptoms and improves overall survival [76,77] . The genus *Blautia* (*Blautia obeum, Blautia hydrogenotrophic*, and *Blautia hansenii*) is beneficial to allo-HSCT [78]. *Blautia coccoides* group (cluster XIVa) was significantly reduced in aGVHD patients, and its abundance was associated with GVHD-related mortality and severity [77,79]. Furthermore, *Faecalibacterium* (*Clostridiaceae, Clostridiales*) and the *Eubacterium limosum* groups were negatively correlated with GVHD [76]. *Peptostreptoccaceae* in *Clostridiales* decreased significantly during GVHD. Therefore, most bacteria under the *Clostridiales* help reduce GVHD and are negatively correlated with severe GVHD development. However, the amplification of *Veillonella parvula* in *Veillonella* and *Rothia mucilaginosa* in *Rothia* was associated with increased GVHD-related mortality [80].

The *Erysipelotrichaceae* of *Erysipelotrichales* under *Erysipelotrichia Erysipelatoclostridium* decreased significantly in the event of GVHD [81].

4.2. Changes of flora in Bacteroidetes

Bacteroides belong to *Bacteroidales* of Bacteroidia and can promote and inhibit intestinal GVHD incidence [82]. *Bacteroides luti*, *Bacteroides thetaiotaomicron*, *Bacteroides ovatus*, and *Bacteroides caccae* were negatively correlated, whereas *Bacteroides dorei* was positively correlated with intestinal GVHD [80].*Bacillus fragilis* showed anti-inflammatory effects in vitro and in vivo, and reduced inflammatory cytokines (TNF-α, IL-1β, IL-6) and increased IL-10 levels in colon tissue [83].It could also improve the development of intestinal GVHD by increasing the gut microbiota diversity, increasing short-chain fatty acids, IL-22, and regulatory T cells [84].Thus, the abundance of *Bacteroides fragilis* in the gut flora was negatively correlated with the severity of GVHD. *Porphyromonadaceae* Obligate anaerobes of the Genus *Barnesiella* protect the intestine by clearing the intestinal vancomycin-resistant Enterococcus (VRE) colonization [85].

4.3. Bacterial changes in other related phyla

Proteobacteria: Amplification of *Enterobacteriales* from γ-Proteobacteria was observed in animal GVHD models [81]. The intestinal inflammation provides a favorable environment for expanding *Enterobacteriaceae* [86]. *Enterobacteriaceae* are effective inflammatory PAMPs and are involved in LPS biosynthesis, promoting Th17 activation, Treg/Th17 imbalance, intestinal inflammation, and GVHD [87]. Therefore, the increased abundance of γ-Proteobacteria, including *Enterobacteriaceae*, is associated with increased GVHD-related mortality. However, the role of *Escherichia coli* (*E. coli*) in the GVHD progression requires further studies. It may produce indole to prevent GVHD [88]. However, a preclinical study found an association between *E. coli* amplification with worse GVHD. Oral polymyxin B can reduce *E. coli* levels and improve GVHD [89]. *Campylobacter* in *Epsilonproteobacteria* was significantly associated with reduced GVHD risk [78].

Verrucomicrobia: *Akkermansia* and *Akkermansia muciniphila* were markedly associated with alleviated GVHD risk [78]. However, previous studies have suggested that *Akkermansia muciniphila* may increase GVHD incidence [29].

Actinobacteria: *Coriobacteriia* is a class of Gram-positive bacteria in *Actinobacteria*, which negatively correlates with the GVHD incidence [90]. *Bifidobacteriaceae* in *Bifidobacteriales* is associated with subsequent GVHD after transplantation [91].

The loss of microbial diversity and the amplification of related pathogens were associated with patient outcomes (such as GVHD severity [77] and related mortality [29], NRM, overall survival [79], and infection risk [65,92]. Therefore, it is clinically possible to reduce GI-GVHD risk by preventing gut microbiota imbalance in allo-HSCT recipients.

5. Effect of antibiotics on the intestinal microflora

Under physiological conditions, the intestinal flora is highly diversified at the microbial species level due to the influence of antibiotics [58], lifestyle [93], diet [94], and geographical environment [95]; however, the total number of microbial genes and the genes involved in the specific metabolic pathway metagenome remain relatively stable among healthy individuals [96]. Research indicates that the intestinal bacteria diversity markedly reduces after allo-HSCT because of antibiotics use and GI-GVHD occurrence [78].

Antibiotics markedly affect gut microbiota composition and the related GI reactions after HSCT. Previous studies have shown that the incidence of GVHD was significantly reduced after complete intestinal purification with high-dose non-absorbable antibiotics [97], But after the current use of immunosuppressors for prevention, the effect of active intestinal decontamination is more complex, leading to the loss of gut microbiota diversity, aggravated impairment of the intestinal barrier and increased GVHD-related mortality [29,57–59]. Penicillin derivatives and carbapenem broad-spectrum antibiotics were independent risk factors for intestinal GVHD [98]. Carbapenem use for >7 days was associated with an increased risk of intestinal GVHD [99]. The cumulative GI-GVHD incidence also notably increased in patients treated with fourth-generation cephalosporin [100]. Narrow-spectrum antibiotic rifaximin helps to maintain intestinal microbial diversity, reduces enterococci proliferation, and increases urinary 3-indolyl sulfate concentration [101]. Therefore, it can reduce GVHD and transplant-related mortality and increase overall survival. Vancomycin is suggested to affect intestinal microbial diversity; however, it does not enhance the GI-GVHD risk or disease recurrence [102,103]. Moreover, it can prevent *Clostridium difficile* infection in allo-HSCT recipients [103]. The use of anaerobic antibiotics for allo-HSCT, such as imipenem-cilastatin, piperacillin-tazobactam, ticarcillin, meropenem, clindamycin, and metronidazole, can significantly decrease intestinal microbial diversity and increase intestinal GVHD-related mortality [29]. Imipenem-cilastatin aggravates GVHD by disrupting the intestinal mucus layer, amplifying the symbiotic bacteria *Akkermansia muciniphila* with mucus degradation ability, and reducing the colonization of Clostridium [29]. However, recently it was indicated that mucin-degrading bacteria stimulate ISC-mediated epithelial development and maintain intestinal homeostasis [104]. Clindamycin can aggravate intestinal GVHD by consuming Clostridium but can help reduce GVHD by oral Clostridium (AIC) [91]. Therefore, exposure to broad-spectrum and/or anaerobically active antibiotics may be a risk factor for GI-GVHD.

6. Recovery of the gut microbiota induced by fecal microbiota transplantation

Fecal microbiota transplantation (FMT) is the process of transferring normal fecal microbes to the patient's gastrointestinal tract to restore its microbiota. It was first used to treat colitis in 1958, and has since rapidly expanded to treat diseases such as intestinal inflammation (ulcerative colitis and Crohn's disease) [105].However, FMT is currently only approved for the treatment of multiple recurrent or refractory *Clostridium difficile* infections (CDI) [106].The steroid-refractory gastrointestinal GVHD faces poor prognosis and limited treatment options [107], but there are numerous experiments showing that FMT can improve GI-GVHD by restoring gut microbiome α diversity, increasing the abundance of Clostridia and Blautia [108,109], increasing peripheral regulatory T cells and ILC3 [110], and also improving progression-free survival in gastrointestinal GVHD [111]. Thus, FMT may serve as a treatment option for steroid-refractory GI-GvHD [112]. In addition, it has been shown that FMT combined with ruxolitinib may be an effective treatment for intestinal steroid-refractory GVHD (SR-GVHD) after HSCT [113]. While safety concerns are the reason for excluding immunocompromised patients from most donor FMT clinical trials, except for the gastrointestinal side effects after FMT, pathogen transmission is the reason that will exclude FMT clinical trials. In the case of severe immunosuppression and intestinal barrier damage after HCT, the introduction of millions of additional bacteria into the gut may translocate microbial organisms to blood flow or even emergence of life-threatening sepsis [114]. Despite the potential of donor FMT as a treatment for steroid-refractory or steroid-dependent GvHD, larger clinical trials are needed to confirm the safety and efficacy of this procedure [109].

7. Effect of microbiota metabolites on the gut

Intestinal microbiota regulates intestinal inflammation and immune response by producing metabolites, thereby alleviating IEC damage and GI-GVHD [115]. *Clostridium leptum* can produce short-chain fatty acids (SCFAs) to reduce intestinal barrier damage and induce immune tolerance [4]. Moreover, the integrity of the epithelial barrier and its function are also maintained by other intestinal or bacterial metabolites, including bile acids (BA), polyamines, and aromatic receptor (AhR) ligands [116]. Therefore, metabolites produced by gut microbiota play an important role in maintaining the intestinal barrier and immune homeostasis.

SCFAs can be used as the main energy source of IECs by up-regulating the tight junction proteins expression, stimulating goblet cells to produce mucins, and promoting IEC histones acetylation [117]. These processes improve the integrity of IEC connectivity and reduce epithelial cell apoptosis and GVHD [115,118]. In addition, SCFAs have multiple effects on immune cells and can enhance host defence without damaging normal tissues [119]. It has a significant anti-inflammatory effect as it promotes the proliferation and differentiation of peripheral Treg [120]and regulates Treg/Th17 balance [121]. It can also activate NLRP3 inflammasome by binding to G protein-coupled receptor 43 (GPR43), GPR41, and GPR109A to phosphorylate ERK, stimulating mitogen-activated protein kinase signal transduction, and promoting the rapid production of chemokines and cytokines, indicating its protective effect on tissue immunity and inflammation [122–124]. Butyrate in SCFAs can inhibit the differentiation, maturation, and function of dendritic cells and macrophages, inhibit the production of IL-12 and interferon-γ (IFN-γ), block NF-kappa B signaling pathway, and promote IL-10 [125], thereby inhibiting GVHD development.

Bile acid exhibits antibacterial properties and affects the survival and growth of the microbial community due to its structural similarity with AMP. Secondary BA metabolized by the microbiota can further affect microbiota composition and regulate host immunity [126,127]. It can induce NLRP3 phosphorylation and ubiquitination through PKA to control inflammation and metabolic disorders [128]. Furthermore, BAs alleviate GVHD severity by reducing antigen-presenting activity and preventing IEC apoptosis without changing the intestinal bacterial composition and graft anti-leukemia effects [129]. The secondary BA metabolite 3β-hydroxydeoxycholic acid can also reduce the immune stimulation characteristics of DCs [130].

Polyamines promote intestinal epithelial renewal, maintain IEC barrier integrity [131], and inhibit monocytes and macrophages activation and pro-inflammatory factors syntheses such as tumor necrosis factor (TNF), IL-1, IL-6, macrophage inflammatory protein-1α (MIP-1α) and MIP-1β [132]. It plays a vital role in balancing autoimmune/inflammation and immunosuppression. Therefore, polyamine levels act as a marker for monitoring immunity and immunotherapy response [133]. In addition, polyamines also promote the maturation of the intestinal and systemic immune systems [134].

Indole and its derivatives are the products of bacterial tryptophan catabolism, which are subsequently metabolized by the liver to 3-indole sulfate (3-IS) and excreted in urine [135]. They are essential for immune homeostasis and act via aromatic hydrocarbon receptor (AhR) signaling [136,137]. Aromatic receptor signaling inhibits T cell-mediated intestinal inflammation by maintaining innate lymphocyte 3 (ILC3) [44] and promotes ILCs to secrete IL-22 to protect intestinal mucosa from infection [138]. It also negatively modulates Th17 development via Stat1 and Stat5. To retain anti-tumor response, indole metabolites limit GI-GVHD-related intestinal inflammation and injury through type I IFN [88]. Additionally, indole and its derivatives also maintain intestinal cells homeostasis, such as inducing epithelial cell proliferation and promoting the synthesis of epithelial tight junction-related proteins, maintaining ISC homeostasis and barrier integrity to protect the intestine from inflammatory damage [139–141]. They also reduce GVHD occurrence and are associated with GVHD-related mortality [142]. In addition, lower 3-IS in urine is associated with increased transplant-related mortality and intestinal GVHD; therefore, it can be used as a biomarker for predicting GVHD [143].

8. Outlook

Patients with GI-GVHD have typical clinical manifestations such as nausea, diarrhea, and abdominal pain [144]. Digestive endoscopy shows glandular dilatation and crypt abscess or occlusion, and histological examination reveals epithelial cell apoptosis, crypt/gland apoptosis, and shedding. Impaired intestinal epithelial tissue, especially Paneth and goblet cells responsible for maintaining the homeostasis between intestinal microbiota and immune response, aggravates GI-GVHD and reduces mucus secretion and AMPs with an epithelial barrier function [25,26] .ISC in the intestinal epithelium are also a major target of GI-GVHD and their damage causes intestinal ecological imbalance or dysbacteriosis and weakened epithelial tissue repair capacity [38]. The imbalance of intestinal flora and bacterial metabolite alterations are related to GVHD risk. During GI-GVDH, maintaining homeostasis of intestinal microbial、Increasing microbial metabolites such as short-chain fatty acids and AMPs、Performing FMT transplantation to restore the diversity of intestinal microflora may restore intestinal epithelial tissue damage. Furthermore, if we can optimize the effects of pretreatment on the gut before allo-HSCT transplantation, reduce the use of broad-spectrum and obligate anaerobic antibiotics to maintain the intestinal bacterial diversity, and promote the proliferation of intestinal Clostridium with microbial metabolites that maintain the intestinal epithelial barrier and intestinal immune homeostasis, this may reduce the incidence of GI-GVHD due to the intestinal microbial imbalance, thus improving patient outcome. However, only a small number of intestinal microorganisms affecting GI-GVHD have been studied. It is believed that with the popularization of high-throughput sequencing technology in the near future, the effect of different intestinal bacteria on GI-GVHD can be more accurately understood, thereby reducing its incidence.

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CRediT authorship contribution statement

Hao Ji: Writing – original draft, Writing – review & editing. **Shuai Feng:** Validation. **Yuan Liu:** Visualization. **Yue Cao:** Writing – review & editing. **HuiQuan Lou:** Writing – review & editing. **Zengzheng Li:** Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare no conflict of interest.

Data availability

No data was used for the research described in the article.

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